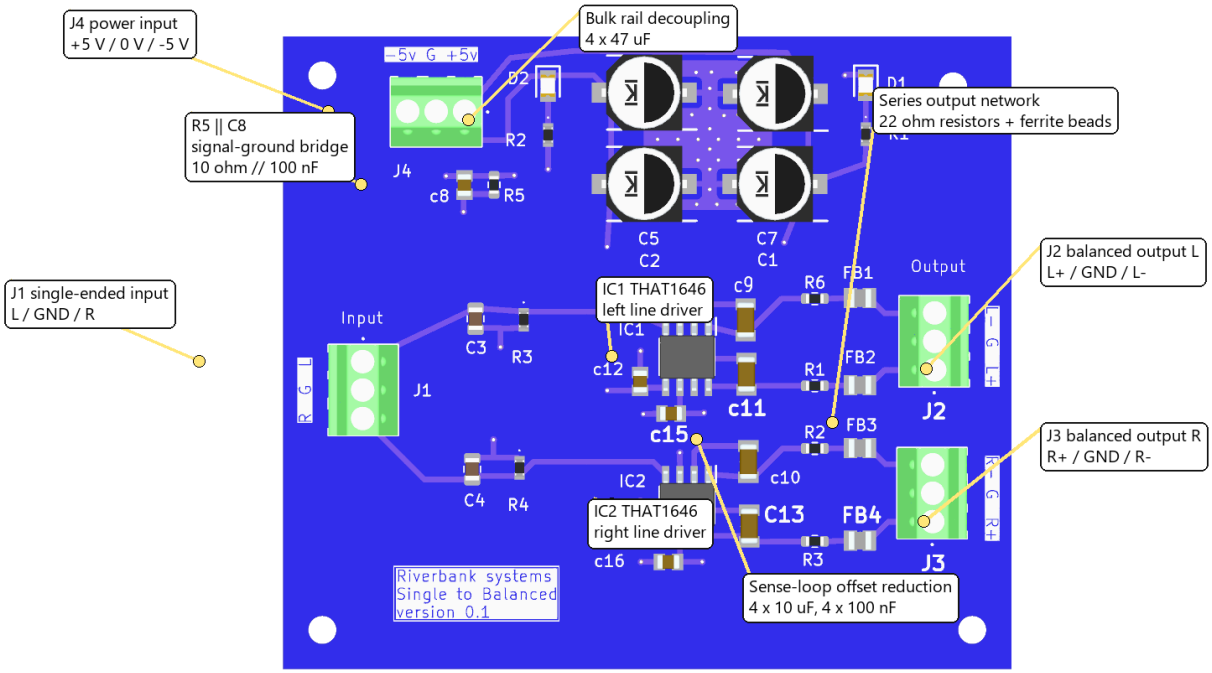
SE to Diff board - Engineering and Service Document

- SE to Diff board: Engineering and Service Document
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SE to Diff board: Engineering and Service Document

Evidence tags: [K] taken from the KiCad schematic, PCB or part metadata; [D] verified from a manufacturer datasheet retrieved for this document; [C] calculated here from [K]/[D]; [I] inferred, not confirmed; [M] needs measurement on hardware. **Nothing in this document was measured on a physical board.**



1. Document information

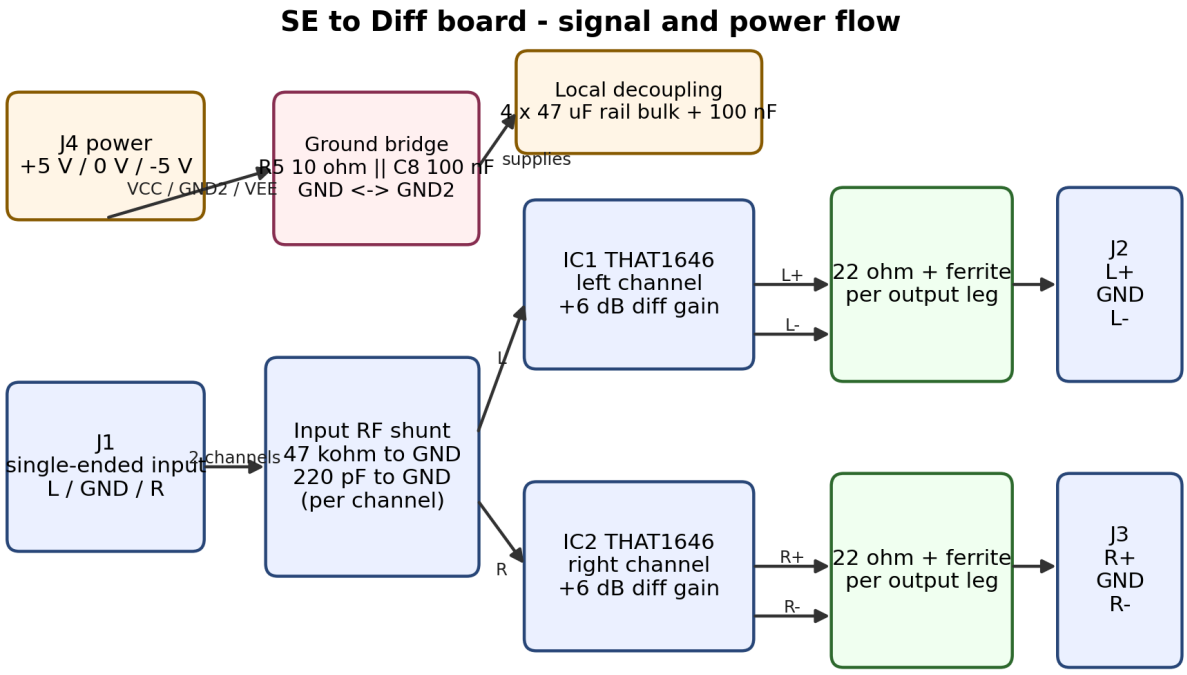
Item	Value
Board	`Riverbank systems / Single to Balanced / version 0.1` silkscreen [K]
Source	`C:\Users\antho\Documents\StreamerDac Project\Documents\kicad\Streamer schematic\ControlBoardTiny\tinyControlBoard\se_to_diff board\se_to_diff board.kicad_sch` and `.kicad_pcb` [K]
KiCad source format	KiCad 8.0 project files, analysed with `kicad-cli` 10.0.6 without modifying the source [K]
Source dates	Schematic last saved 2026-01-28 21:25:51; PCB last saved 2026-01-29 19:20:37 [K]
Revision / title block	Schematic title, date and revision are blank [K]
Related	Open questions; superseded subsystem note: Single-Ended to Balanced Line Driver

This document supersedes the older subsystem-only note for service work on this PCB. One important correction is that the board does **not** present a 47 kOhm line input: the THAT1646 input impedance dominates, so the effective input load is about **4.52 kOhm per channel** [C][D][K].

2. Overview

The SE to Diff board is a **two-channel single-ended to balanced line-driver board** [K]. Each channel uses one **THAT1646S08-U** balanced line-driver IC (IC1 left, IC2 right) powered from a **bipolar +5 V / -5 V supply** brought in on J4 as **VCC / GND2 / VEE** [K]. A single-ended stereo input enters on J1, each channel is conditioned by a small RF-shunt network, then converted to balanced outputs that leave on J2 and J3 through **22 ohm series resistors** and **ferrite beads** on every output leg [K].

The board has two different ground nets: **GND2** is the supply mid-point used by the IC supply decoupling, while **GND** is the signal ground used by J1, J2, J3 and IC pin 3. They are linked only by **R5 10 ohm in parallel with C8 100 nF** [K]. This is a deliberate ground-break network, not a solid copper short [K].



Not established from these files: the exact source impedance of the upstream DAC/output stage, the intended cable/shield strategy on J2/J3 pin 2, and the maximum system-level balanced output level required by the rest of the product [M].

3. Specifications

Parameter	Value	Tag
PCB	65.0 mm x 56.5 mm, 4 copper layers, 1.6 mm board thickness	[K]
Assembly side	All fitted parts are on the top side	[K]
Mounting	4 x 2.5 mm mounting holes	[K]
Channels	2 (left and right)	[K]
Supply input	J4 pin 1 = VCC, pin 2 = GND2, pin 3 = VEE	[K]
Nominal supply implied by schematic text	+5 V VCC and -5 V VEE relative to GND2	[K]
Driver IC	THAT1646S08-U balanced line driver, SO-8	[K][D]
Driver supply range	-4 V to +18 V allowed; absolute maximum +/-20 V	[D]
Driver operating temperature	-40 C to +85 C	[D]
Differential gain	+6.0 dB typ. (factor 2)	[D]
Driver input impedance	5.0 kOhm typ. per channel	[D]
Added input shunt	47 kOhm to GND and 220 pF to GND per channel	[K]
Effective input impedance seen at J1	about 4.52 kOhm per channel (`5 kOhm	
Small-signal bandwidth of THAT1646	10 MHz typ.	[D]
Datasheet output noise	-101 dBu, 22 Hz to 20 kHz, balanced, measured at +/-18 V test conditions	[D]
Datasheet THD+N	0.0007 % typ. at 1 kHz equivalent test band, 10 Vrms into 600 ohm balanced, measured at +/-18 V test conditions	[D]

Parameter	Value	Tag
Output impedance of IC	50 ohm typ. per leg	[D]
Added series resistance	22 ohm on each output leg, plus ferrite bead	[K]
Approximate output impedance at audio frequencies	about 72 ohm per leg, excluding ferrite bead HF behaviour	[C][D] [K]
Approximate maximum differential output on this +5 V / -5 V board	typ. 3.92 Vrms (+14.1 dBu); guaranteed by no-load swing limits at least 2.97 Vrms (+11.7 dBu)	[C][D]
Approximate maximum input before differential clipping	typ. 1.96 Vrms; guaranteed at least 1.48 Vrms, because gain is +6 dB	[C][D]
IC quiescent current	4.9 mA typ. to 5.75 mA max each, unloaded	[D]
IC-only quiescent dissipation at +5 V / -5 V	about 57.5 mW per IC, 115 mW total for two ICs	[C][D]
Connector family	Phoenix Contact 1725669, 3-way, 2.54 mm pitch screw terminal block; KiCad metadata states 6 A nominal and 160 V (III/2)	[K]

Notes on the table above:

- The **THD+N and noise values are datasheet figures taken at +/-18 V**, not measured results from this +5 V / -5 V board [D][M].
- The board should be treated as a **low-impedance audio driver stage**, not a high-impedance line input, because the THAT1646 input itself is only about 5 kOhm [D].
- Thermal margin is good in normal use: 57.5 mW in a 150 C/W SO-8 package implies only about **8.6 C** self-heating per IC before PCB heat-spreading is counted [C][D].

4. Connectors, controls, indicators and test points

4.1 Connectors

Ref	Function	Pin 1	Pin 2	Pin 3
J1	Single-ended stereo input	`/in L`	`GND`	`/in R`
J2	Left balanced output	`/Sout L +`	`GND`	`/Sout L -`
J3	Right balanced output	`/Sout R +`	`GND`	`/Sout R -`
J4	Power input	`VCC`	`GND2`	`VEE`

Important service note: **J1/J2/J3 use `GND`, while J4 uses `GND2`** [K]. These are not directly shorted on the PCB; they are linked only through R5 and C8 [K].

4.2 Indicators

Ref	Function
D1 + R1	Positive-rail indicator from VCC to GND2 [K]
D2 + R2	Negative-rail indicator from VEE to GND2 [K]

The source files do not state LED forward voltage or target brightness. If the schematic value **3 kOhm** is correct, LED current is expected to be low [K][I].

4.3 Controls and test points

- No switches, jumpers or adjustable parts are fitted [K].
- No dedicated test points are provided [K]. Service measurements must be taken at connector pins or IC pins [K].

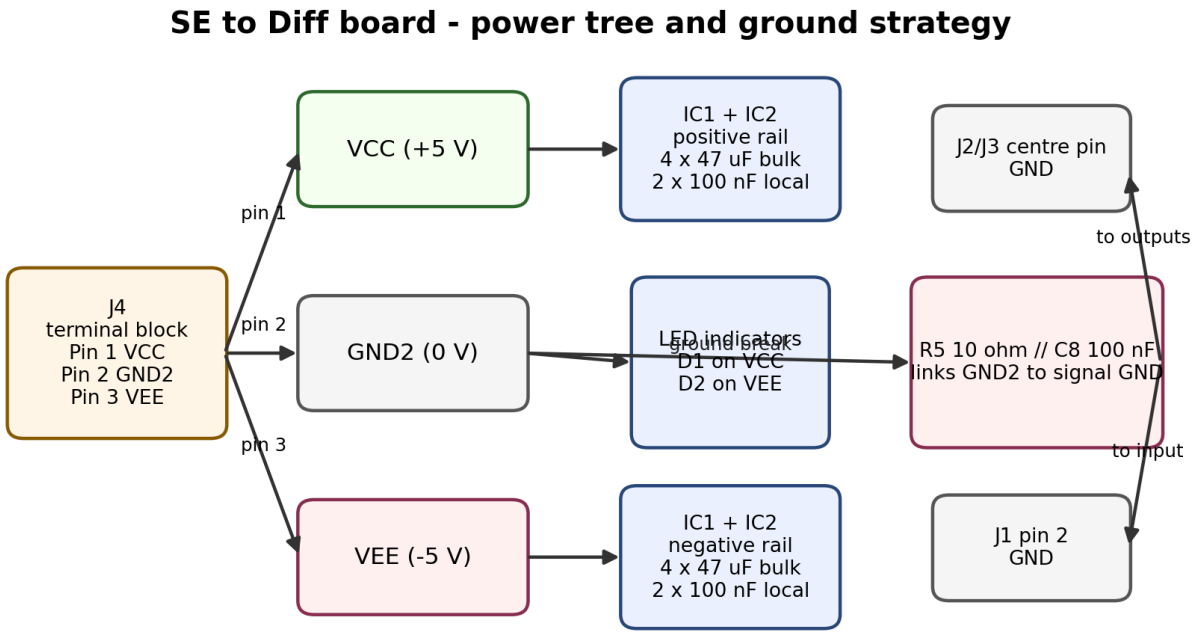
- Useful service nodes are J4 pins 1/2/3, J1 pin 2, J2 pin 2, J3 pin 2, IC1/IC2 pins 5 and 6, and the output nets before and after the ferrite beads [K].

5. Functional description

5.1 Power and grounding

J4 feeds the board with **+5 V, 0 V, -5 V** on **VCC, GND2, VEE** respectively [K]. The two THAT1646 devices run directly from these rails [K]. VCC has bulk capacitors **C1 and C7 (47 uF each)**, VEE has **C2 and C5 (47 uF each)**, and each IC has local **100 nF** rail bypass capacitors (**C12/C15** for IC1, **C14/C16** for IC2) returned to **GND2** [K].

The supply ground **GND2** is linked to the signal ground **GND** only by **R5 10 ohm || C8 100 nF** [K]. The impedance of that bridge is therefore about **10 ohm at low audio frequencies** and about **8.88 ohm at 20 kHz** [C]. This is useful as a loop-break network, but it also means any return or shield current flowing between `GND` and `GND2` will create a small voltage difference instead of a hard short [C][I].



5.2 Input network

Each input channel consists of:

- one **47 kOhm** resistor to GND (R3 left, R4 right) [K];
- one **220 pF** capacitor to GND (C3 left, C4 right) [K];
- direct connection into pin 4 (`IN`) of the THAT1646 [K].

The **220 pF capacitor is an RF shunt**, not an audio low-pass set by 47 kOhm. The audio pole depends on the source impedance feeding J1. For example, the pole would be about **1.21 MHz with a 600 ohm source** and about **723 kHz with a 1 kOhm source** [C].

The important practical result is instead the **heavy input loading**: the THAT1646 itself is about **5 kOhm**, so J1 loads each source with about **4.52 kOhm** in total [C][D][K]. The THAT datasheet explicitly recommends a **low-impedance source, preferably directly from op-amp outputs** [D].

5.3 Differential driver stage

IC1 and IC2 are wired very closely to the datasheet's **Figure 5** application for the THAT1646 with common-mode offset reduction [D][K]:

- pin 1 = `OUT-` [K]
- pin 2 = `SNS-`, AC-coupled to `OUT-` through **10 uF** (C9 left, C10 right) [K]
- pin 3 = `GND` [K]
- pin 4 = `IN` [K]
- pin 5 = `VEE` [K]
- pin 6 = `VCC` [K]
- pin 7 = `SNS+`, AC-coupled to `OUT+` through **10 uF** (C11 left, C13 right) [K]
- pin 8 = `OUT+` [K]

With these sense capacitors fitted, the datasheet's common-mode output offset limit improves from **up to +/-250 mV** to about **+/-15 mV max.** [D].

5.4 Output network

Each output leg goes through a **22 ohm series resistor** and then a **Murata NFZ2MSD101SZ10L ferrite bead** before reaching J2 or J3 [K]. The mapping is:

- Left channel: `R6` and `22R1`, then `FB1` and `FB2` [K]
- Right channel: `22R2` and `22R3`, then `FB3` and `FB4` [K]

This follows the **spirit** of the THAT1646 datasheet's RFI-protection guidance, but **not the full Figure 8 protection network** [D][K]. The PCB does **not** include the datasheet's recommended **100 pF shunts to chassis** or the optional **1N4004 rail-clamp diodes** for 48 V phantom-power abuse [D][K]. If J2/J3 can ever reach studio wiring, XLR harnesses or other field cabling, that omission should be reviewed [I].

5.5 Level and headroom implications

The THAT1646 has **+6 dB differential gain** [D]. On +5 V / -5 V rails, output headroom becomes the dominant limit. Using the datasheet output-swing limits, this board can deliver about:

- **3.92 Vrms differential typ.** (+14.1 dBu) [C][D]
- **2.97 Vrms differential worst-case guaranteed by no-load swing limits** (+11.7 dBu) [C][D]

That means a **2 Vrms single-ended source** would ask the board for **4 Vrms differential output**. This is slightly above the typical calculated swing and clearly above the guaranteed worst-case swing [C][D]. In other words, a modern 2 Vrms consumer DAC may be at or beyond the board's clean headroom limit [C][I].

6. Assembly and commissioning

No build or power-up results are claimed here; the following is a suggested service procedure only [M].

1. **Visual inspection** [M]
   - Confirm J1-J4 are the correct 3-way terminal blocks and are not rotated [K].
   - Check polarity of **C1, C2, C5, C7** and LEDs **D1, D2** [K].
   - Check that **IC1** and **IC2** pin 1 orientation matches the PCB footprint [K].
2. **Unpowered resistance checks** [M]
   - J4 pin 1 to pin 2 and pin 3 to pin 2 should not be short circuits [K].
   - J1 pin 2 to J4 pin 2 should read about **10 ohm DC** through R5, not a direct short [K][C].
3. **First power application** [M]
   - Use a current-limited bench supply that provides **+5 V, 0 V and -5 V**.
   - Check J4 pin 1 to pin 2 = **+5 V** and pin 3 to pin 2 = **-5 V** [K].
   - Check IC1/IC2 pin 6 to GND2 = **+5 V** and pin 5 to GND2 = **-5 V** [K].
   - D1 and D2 should light if both rails and LED polarities are correct [K][I].

4. **Signal proving** [M]
    - Inject a low-level sine wave into J1 pin 1 or pin 3 with J1 pin 2 as signal ground.
    - Verify that the corresponding output connector presents two equal-amplitude, opposite-phase signals on pins 1 and 3 [D][K].
    - Keep the input below about **1.5 Vrms** until actual headroom is measured on hardware [C][M].

7. Testing and troubleshooting

Symptom	Likely cause	Check
Both channels dead, D1 and D2 off	Missing or reversed +5 V / -5 V feed	J4 pin voltages relative to J4 pin 2 [K]
Both channels dead, LEDs on	IC supply not reaching pins 5/6, or GND/GND2 not as expected	Measure IC pin 6 to GND2, pin 5 to GND2, and J1/J2/J3 pin 2 to J4 pin 2 resistance [K]
One channel dead	Fault in one THAT1646, one input network, or one output network	Compare IC1 against IC2, then trace from J1 to IC input and from IC output to J2/J3 [K]
One output leg dead	Open 22 ohm resistor, ferrite bead or connector pin	Check continuity through R6/22R1/22R2/22R3 and FB1-FB4 [K]
Distortion with an apparently normal line-level source	Board headroom exceeded on +5 V / -5 V rails	Check input level; clean differential output is only about 3.0 to 3.9 Vrms expected [C][D]
Source device becomes heavily loaded or level drops	Upstream source cannot drive about 4.5 kOhm	Measure source behaviour into J1; review whether the source is truly low impedance [C][D]
Hum, buzz or shield-current problems	`GND` and `GND2` are only linked by 10 ohm // 100 nF	Check where cable shields are terminated and whether the system expects a direct 0 V bond [K][I]
Large DC at J2/J3 outputs	Wrong or open sense capacitors C9/C10/C11/C13, or wrong IC wiring	Compare to datasheet Figure 5 topology and measure DC on both output legs [D][K][M]

8. Repair and rework

- Replace **IC1/IC2 only with the correct THAT1646S08-U** unless an alternative has been electrically re-qualified for this board [K][D]. The datasheet notes pin-compatibility with some other line drivers, but this document does not approve substitutes [D].
- Keep **C9/C10/C11/C13 (10 uF)** and the **100 nF local bypass capacitors** physically close to the IC pins, as in the existing layout [K].
- If a ferrite bead or 22 ohm resistor is damaged, replace like-for-like; do not hard-short around the network without a noise/EMC review [K][I].
- Do not short **GND** and **GND2** permanently during rework unless the full system grounding strategy has been re-reviewed [K][I].
- After any work on the supply area, repeat the section 6 resistance and rail-voltage checks before reconnecting external audio equipment [M].

9. BOM reference

The table below is a service reference, not a purchasing release. **Several resistor manufacturer fields in the KiCad source are inconsistent with the schematic values** and should not be trusted until the BOM metadata is corrected [K].

Ref(s)	Qty	Value / part information	Notes
IC1, IC2	2	THAT 1646S08-U	Balanced line drivers [K][D]
J1-J4	4	Phoenix Contact 1725669	3-way, 2.54 mm terminal blocks [K]
FB1-FB4	4	Murata NFZ2MSD101SZ10L	Output ferrite beads [K]

Ref(s)	Qty	Value / part information	Notes
C1, C2, C5, C7	4	47 uF, KEMET EDT476M016S9HAA	Rail bulk capacitors [K]
C9, C10, C11, C13	4	10 uF, Kyocera AVX 12063C106KAT2A	Sense-loop capacitors [K]
C8, C12, C14, C15, C16	5	100 nF, 0805YC104M4T2A	Ground bridge and local decoupling [K]
C3, C4	2	220 pF, 0805	Input RF shunts [K]
R1, R2	2	3 kOhm schematic value	LED resistors; KiCad MPN field appears stale/wrong [K]
R3, R4	2	47 kOhm schematic value	Input shunts; KiCad MPN field appears stale/wrong [K]
R5	1	10 ohm	GND-to-GND2 bridge resistor [K]
R6, 22R1, 22R2, 22R3	4	22 ohm	Output series resistors [K]
D1, D2	2	LS R976 red LED	Rail indicators [K]
H1-H4	4	2.5 mm mounting holes	Mechanical only [K]

10. Limitations, risks and open points

Design-check findings from the source files and datasheet review:

1. **This board is easy to overdrive from modern 2 Vrms sources.** With +6 dB gain and only +5 V / -5 V rails, calculated clean differential output is roughly **+11.7 to +14.1 dBu** [C][D]. A 2 Vrms input asks for about **4 Vrms differential output**, which is at or beyond that range [C][D].
2. **The input is low impedance.** The effective load is only about **4.52 kOhm/channel**, not 47 kOhm [C][D][K]. Upstream circuitry should therefore be an active low-impedance source, not a passive attenuator or a lightly loaded DAC output [D][I].
3. **Ground strategy must be intentional.** J1/J2/J3 use `GND`, J4 uses `GND2`, and the only bridge is **10 ohm || 100 nF** [K]. That can help loop control, but it can also create audible or EMC-related ground voltage if cable shield currents are significant [C][I].
4. **The board omits the full datasheet protection network.** The THAT1646 datasheet shows optional **100 pF-to-chassis** RFI shunts and **1N4004** rail clamps for phantom-power abuse; this PCB has neither [D][K].
5. **KiCad resistor metadata is inconsistent.** R1-R4 all carry the same Panasonic **300 ohm** part metadata even though the schematic values are **3 kOhm** and **47 kOhm** [K]. This is a documentation/BOM risk during servicing.
6. **Connector voltage/current figures were available only from KiCad part metadata in this review.** Direct automated retrieval of the Phoenix product page was blocked, so those connector ratings should be rechecked manually before formal release documentation [K][M].
7. **No physical measurements exist yet.** Actual headroom, output noise, LED current, shield-current behaviour and DC offsets on built boards remain to be verified [M].

The superseded subsystem note is mostly consistent with the KiCad source, but it does not mention findings 1, 2, 5 or 6 above, and its wording that the `47 kOhm // 220 pF` network "sets the input impedance" is incomplete for service use because the THAT1646 input dominates the load [K][D][C].

11. References and revision history

References

1. KiCad source files for `se_to_diff board.kicad_sch` and `se_to_diff board.kicad_pcb` [K].
2. THAT Corporation, **THAT 1606, 1646 Balanced Line Driver ICs**, Document 600078 Rev. 05 [D].
3. Existing subsystem note: Single-Ended to Balanced Line Driver [K].

Revision history

Revision	Date	Notes
A	2026-10-11	First engineering and service issue created from KiCad source review, ERC/DRC runs, datasheet review and desk calculations.